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Experiment-37

Algorithm

Step 1: Start.

Step 2: Declare three separate functions:

- `toUppercase()`
- `reverseString()`
- `checkPalindrome()`

Step 3: Display the menu:

1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit

Step 4: Read the user's choice.

Step 5: Use `switch` to perform the selected operation.

Case 1: Convert Lowercase to Uppercase

1. Read the string.
2. Start from the first character.
3. Check each character until `'\0'` is reached.
4. If the character is between `'a'` and `'z'`, subtract 32 from its ASCII value.
5. Display the uppercase string.
6. Return to the menu.

Case 2: Reverse a String

1. Read the string.
2. Find the length of the string using a loop.
3. Start from the last character.
4. Print each character up to the first character.
5. Display the reversed string.
6. Return to the menu.

Case 3: Check Palindrome

1. Read the string.
2. Find the length of the string using a loop.
3. Set `flag = 1`.
4. Compare the first character with the last character, second with second-last, and so on.
5. If any pair is different, set `flag = 0` and stop checking.
6. If `flag = 1`, display **"The string is a palindrome."**
7. Otherwise, display **"The string is not a palindrome."**
8. Return to the menu.

Case 4: Exit

1. Display "Exiting program...".
2. Stop the program.

Step 6: If the choice is not between 1 and 4, display "Invalid choice!".

Step 7: Repeat the menu until the user selects 4.

Step 8: Stop.

Test Case Table

Test Case	Choice	Input String	Expected Output
TC1	1	hello	HELLO
TC2	1	good morning	GOOD MORNING
TC3	2	computer	retupmoc
TC4	2	hello world	dlrow olleh
TC5	3	madam	The string is a palindrome.
TC6	3	level	The string is a palindrome.
TC7	3	hello	The string is not a palindrome.
TC8	5	—	Invalid choice!
TC9	4	—	Exiting program...

C Program

```
#include <stdio.h>
```

```
void toUppercase();  
void reverseString();  
void checkPalindrome();
```

```
int main()  
{
```

```

int choice;

do
{
    printf("\n--- MENU ---\n");
    printf("1. Convert lowercase string to uppercase\n");
    printf("2. Reverse a string\n");
    printf("3. Check if a string is a palindrome\n");
    printf("4. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            toUppercase();
            break;

        case 2:
            reverseString();
            break;

        case 3:
            checkPalindrome();
            break;

        case 4:
            printf("Exiting program...\n");
            break;

        default:
            printf("Invalid choice!\n");
    }

} while(choice != 4);

return 0;
}

```

```

void toUppercase()
{
    char str[100];
    int i;

    printf("Enter a lowercase string: ");
    scanf(" %[^\\n]", str);

    for(i = 0; str[i] != '\\0'; i++)

```

```

    {
        if(str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - 32;
        }
    }

    printf("Uppercase string: %s\n", str);
}

```

```

void reverseString()
{
    char str[100];
    int i, length = 0;

    printf("Enter a string: ");
    scanf("%[^\\n]", str);

    while(str[length] != '\\0')
    {
        length++;
    }

    printf("Reversed string: ");

    for(i = length - 1; i >= 0; i--)
    {
        printf("%c", str[i]);
    }

    printf("\\n");
}

```

```

void checkPalindrome()
{
    char str[100];
    int i, length = 0;
    int flag = 1;

    printf("Enter a string: ");
    scanf("%[^\\n]", str);

    while(str[length] != '\\0')
    {
        length++;
    }

    for(i = 0; i < length / 2; i++)

```

```
{
    if(str[i] != str[length - 1 - i])
    {
        flag = 0;
        break;
    }
}

if(flag == 1)
    printf("The string is a palindrome.\n");
else
    printf("The string is not a palindrome.\n");
}
```